

Strategic Under-Specification: A Stackelberg Game Between User and Assistant

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Abstract

Users of coding assistants routinely under-specify their intent, and the assistant’s policy for handling missing information—guess silently, or ask a clarifying question—is itself a strategic choice that users respond to. We formalize this as a Stackelberg game: the assistant commits to an ask/guess policy, and a population of users with privately known, heterogeneous specification costs best-respond. We first solve a decision-theoretic benchmark in which the policy is exogenous. Here, improving the assistant’s raw guessing accuracy alone cannot make users worse off, even though it induces them to specify less—an immediate envelope-theorem consequence of letting a single agent optimize freely against an improved technology, but one worth stating explicitly because it is the benchmark against which the paper’s actual moral-hazard channel (a gap between believed and true policy) should be measured; naive “better guessing causes worse specs and worse outcomes” stories conflate the two. We then close the loop and solve for the assistant’s own optimal policy under two regimes: a *pooling* policy restricted to a single population-wide ask rate, and a *screening* policy that can condition on the user’s declared specification. We show that an unbiased (welfare-aligned) assistant achieves full efficiency under screening despite users’ private information—incentive compatibility is free when principal and agent share an objective—so all screening-regime inefficiency is attributable to misalignment between the assistant’s effective objective and true user welfare, not to private information per se. Pooling, by contrast, loses efficiency from restricted instruments even when unbiased. We characterize both equilibria in closed or near-closed form under a tractable linear-risk specification, derive testable comparative statics, and lay out a simulation and human-study design to test them.

1 Introduction

A user typing a coding request faces a choice: spend effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user’s specification effort depends on what they expect the assistant to do with silence, and the assistant’s ask/guess policy is chosen (explicitly, or as an emergent property of training) with some implicit model of how users will respond. Section 2 treats the assistant’s policy as exogenous and solves the user’s problem against it—a one-sided optimization, useful as a benchmark but not yet a game. Sections 4–7 close the loop: the assistant is a Stackelberg leader that commits to a policy anticipating the population’s best response, and we solve for that policy under two different instrument restrictions, then compare them.

2 Environment and the Decision-Theoretic Benchmark

2.1 Setup

A task has k binary, independent, intent-relevant attributes $\theta = (\theta_1, \dots, \theta_k) \in \{0, 1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in \{0, \dots, k\}$ and truthfully reveals m attributes.¹ Specifying m attributes costs $c(m)$, with $c(0) = 0$, $c' > 0$, $c'' \geq 0$.

For each of the $k - m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0, 1)$. We model the assistant's policy at this stage as a uniform ask rate $a \in [0, 1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a, g) = a + (1 - a)g$. Task success requires *all* k attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user's payoff is

$$U(m; a, g) = V q(a, g)^{k-m} - c(m) - a(k - m)\kappa,$$

with $V > 0$ the value of success and $\kappa > 0$ the user's cost of answering one clarifying question. We treat m as continuous on $[0, k]$ for tractability.

2.2 Proposition 1: accuracy alone does not create moral hazard

Proposition 1. *Fix a . Let $U^*(g) = \max_{m \in [0, k]} U(m; a, g)$. Then U^* is weakly increasing in g .*

Proof. By the envelope theorem, at the optimum $m^*(g)$, $dU^*/dg = \partial U / \partial g|_{m=m^*(g)} = V(k - m^*)q^{k-m^*-1}(1 - a) \geq 0$. The envelope theorem lets us ignore dm^*/dg : because m^* is chosen optimally, its first-order response to g has no first-order effect on U^* . $\square$

Assumption 1 (Regularity). *In the relevant range of m , $(k - m) \ln q(a, g) \leq -1$.*

Remark 1 (Interpreting Assumption 1, and what remains unchecked). *The condition has no closed-form economic reading on its own; it is a sufficient condition for the single-crossing property used in Corollary 1 and Proposition 2 via Topkis's theorem, not a claim about primitives directly. Loosely, since $\ln q \approx -(1 - q)$ for q near 1, the condition is close to $(k - m)(1 - q) \geq 1$, i.e., the expected number of wrong attributes under the current policy is at least one. A numerical audit over $k \in [5, 15]$, $q \in [0.70, 0.95]$, and $\kappa \in [0.20, 2.00]$ (726 grid points, evaluated at each point's own optimizing m^*) finds the condition holds in only 10.1% of this range, failing in 100% of points with $\kappa \leq 0.60$ (users specify almost everything, so $k - m^* \rightarrow 0$) and in effectively all points with $q \geq 0.90$ (guessing is accurate enough that $|\ln q|$ is too small for the bound to bind within $k \leq 15$). This is precisely the regime of a capable, low-friction assistant, arguably the empirically relevant case. Two things follow. First, this is a statement about a sufficient condition, not about the conclusion itself: Corollary 1 and Proposition 2 could still hold by a different argument, or could simply fail, outside this region, and the audit above does not distinguish these possibilities, since it only evaluates whether the sufficient condition holds, not whether $m^*(\kappa)$ or $m^*(g)$ actually lose monotonicity when it doesn't. Whether monotonicity survives numerically outside the 10.1% region is the immediate next check, and until it is done, Corollary 1 and Proposition 2 should be read as formally established only inside that region rather than across the paper's full stated working range.*

¹We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

Corollary 1. *Under Assumption 1, $m^*(g)$ is weakly decreasing in g (decreasing differences in (m, g) ; Topkis’s theorem).*

So, holding the assistant’s policy fixed, users specify less as the assistant gets better at guessing, yet are never worse off for it. We flag explicitly that Proposition 1 is a low bar: it is a direct application of the envelope theorem to a single agent optimizing freely against a technology parameter that enters its payoff with non-negative derivative, and would hold in essentially any model of this form. Its purpose here is not to be a surprising finding but to be the correct benchmark against which any claimed moral hazard should be measured, so that Proposition 2 below can be read as identifying the actual, non-trivial source of harm. It isolates the question this paper is really about: what happens once the policy a is no longer exogenous, but chosen by a strategic assistant that anticipates this response?

3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant’s ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

3.1 Cost externality

If a third party (a reviewer, downstream production systems) bears part of the cost of a wrong guess, the user’s private (κ, V) understate the social values, and the user under-specifies relative to the social optimum at any fixed g —worsening, by Corollary 1, as g rises. Standard; not our focus.

3.2 Miscalibrated belief updating under silent failure

If wrong guesses are not always detected, and detection probability $d(m)$ is increasing in m , users infer g from experience rather than observing it, and under-specifying users are also less likely to catch their own errors—an upward-biased $\hat{g}$ that reinforces under-specification. This is a dynamic learning story rather than a static equilibrium result; we return to it in the empirical design (prediction P5).

3.3 Belief about the assistant’s policy

Let users be *sophisticated* (correctly infer the assistant’s true ask rate a_A) or *naive* (believe the assistant follows the welfare-optimal rate $a^\dagger \geq a_A$).

Proposition 2. *Under Assumption 1, naive users choose $m_{naive}^* \leq m_{soph}^*$ and $U(m_{naive}^*; a_A, g) < U(m_{soph}^*; a_A, g)$: naive users under-specify and are strictly worse off than sophisticated users, holding raw guessing accuracy g fixed throughout.*

Proof. Corollary 1 gives $m^*(\cdot)$ non-increasing in its accuracy argument; m_{soph}^* is optimal against the true $q(a_A, g)$, so any other m , including m_{naive}^* , delivers strictly lower $U(\cdot; a_A, g)$ whenever it differs. $\square$

This is the source of genuine moral hazard identified so far: not raw accuracy, but an *unobserved gap between the user’s belief about the assistant’s policy and the assistant’s true policy*. Section 5 solves for what that true policy a_A actually is when the assistant is itself a strategic, possibly biased, Stackelberg leader—so Proposition 2 can be instantiated with a derived a_A rather than an assumed one.

4 The Assistant's Policy as a Stackelberg Game

4.1 General formulation

The assistant (leader) commits ex ante to a policy π before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m; \kappa) = \kappa m^2/2$, and best-responds $m^*(\kappa; \pi) = \arg \max_m U(m; \pi, \kappa)$. The leader's own objective Π need not equal U : we write

$$\Pi_\kappa(m, a) = \mu_A U(m, a; \kappa) - (\lambda_A - c_Q) a (k - m),$$

where $\mu_A \in (0, 1]$ discounts the leader's weight on true (downstream) user welfare, c_Q is the user's true cost of answering one clarifying question, and λ_A is the leader's own *perceived* per-question friction cost, which may differ from c_Q (e.g., a training signal that penalizes extra turns more than users actually mind answering a direct question). The leader is *unbiased* iff $\mu_A = 1$ and $\lambda_A = c_Q$, in which case $\Pi_\kappa \equiv U(\cdot; \kappa)$ exactly. The leader solves $\pi^* = \arg \max_\pi \mathbb{E}_{\kappa \sim F} [\Pi_\kappa(m^*(\kappa; \pi), \pi)]$, a bilevel program.

Remark 2 (Existence). *For Model I below ($\pi = a \in [0, 1]$, a compact scalar policy), $m^*(\kappa; a)$ is continuous in a (strict concavity of U in m plus the implicit function theorem), so $\mathbb{E}_\kappa[\Pi_\kappa]$ is continuous in a and a maximizer exists on $[0, 1]$ by Weierstrass's theorem. For Model II below (a finite menu over two types), the feasible set is a compact subset of $\mathbb{R}^4$ cut out by linear IC/IR constraints and a continuous objective, so a maximizer again exists by Weierstrass's theorem; no ironing or ex-ante randomization is needed with only two types.*

For tractability in what follows we replace the exact conjunctive success probability q^{k-m} with its first-order (small-risk) approximation around $q \rightarrow 1$: expected loss $\approx L(1-q)(k-m)$, where L is the loss from a single unresolved-and-incorrect attribute. (This linear-risk form is already implicit in the stakes notation $L(s)$ used informally in earlier drafts of this model; here we make it explicit and load-bearing.) With $\Lambda \equiv L(1-g)$, the user's payoff becomes

$$U(m, a; \kappa) = V - \Lambda(1-a)(k-m) - \frac{\kappa}{2} m^2 - a c_Q (k-m).$$

4.2 The first-best benchmark

Before analyzing either regime, characterize the fully-informed, fully-aligned optimum: $(m^{FB}(\kappa), a^{FB}(\kappa)) = \arg \max_{m,a} U(m, a; \kappa)$.

Proposition 3 (First-best is a threshold rule in κ). *U is linear in a for fixed m , so $a^{FB}(\kappa) \in \{0, 1\}$. Define $\kappa^* = (\Lambda + c_Q)/(2k)$. If $\Lambda > c_Q$, then $a^{FB}(\kappa) = 1$ (always ask) for $\kappa > \kappa^*$ and $a^{FB}(\kappa) = 0$ (never ask, guess) for $\kappa < \kappa^*$.*

Proof. At $a = 1$: $m^*(\kappa; 1) = c_Q/\kappa$ and $U(m^*, 1; \kappa) = V - c_Q k + c_Q^2/(2\kappa)$. At $a = 0$: $m^*(\kappa; 0) = \Lambda/\kappa$ and $U(m^*, 0; \kappa) = V - \Lambda k + \Lambda^2/(2\kappa)$. Subtracting,

$$U(a=1) - U(a=0) = [\Lambda - c_Q] \left(k - \frac{\Lambda + c_Q}{2\kappa} \right),$$

which is positive iff $\Lambda > c_Q$ and $\kappa > (\Lambda + c_Q)/(2k) = \kappa^*$. $\square$

This is a clean, closed-form, type-dependent first best: high-cost (terse) users, who cannot cheaply self-provide detail, should always be asked; low-cost (verbose) users, who can cheaply write enough that the small residual is safe to guess, should never be asked. This threshold rule replaces and generalizes the stakes-weighted threshold from earlier drafts of this model (that result is the special case obtained by varying $L(s)$ at fixed κ instead of varying κ at fixed L).

5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0, 1]$ —it cannot condition on anything user-specific. This is the natural benchmark for a policy that is fixed at training time and cannot observe individual users.

5.1 User best response

The FOC $\partial U / \partial m = \Lambda(1 - a) - \kappa m + ac_Q = 0$ gives, with $\beta(\kappa) = \Lambda/\kappa$ and $\gamma(\kappa) = (\Lambda - c_Q)/\kappa$,

$$m^*(\kappa; a) = \beta(\kappa) - a\gamma(\kappa), \quad \frac{\partial m^*}{\partial a} = -\gamma(\kappa) = \frac{c_Q - \Lambda}{\kappa}.$$

Remark 3 (Non-monotone effect of ask rate on specification). *Unlike Corollary 1, the sign of $\partial m^* / \partial a$ here does not depend on type: it is positive for every type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that holds, a higher population ask rate induces users to specify more, not less—they pre-empt costly clarifying questions by writing the answer down themselves. This is a genuinely new force relative to Section 2, where a only entered through accuracy, and is a distinct, directly testable prediction (P1' below).*

5.2 The leader's problem

Let $R(a, \kappa) = k - m^*(\kappa; a)$. The leader's per-type payoff is $\Pi_\kappa(a) = \mu_A V - C(a)R(a, \kappa)$ with $C(a) = \mu_A \Lambda + a(\lambda_A - \mu_A \Lambda)$. Averaging over F with $\bar{R}(a) = \bar{R}_0 + a\bar{\gamma}$ (where $\bar{R}_0 = k - \mathbb{E}[\beta]$, $\bar{\gamma} = \mathbb{E}[\gamma]$), $\Pi(a)$ is quadratic in a :

$$\Pi(a) = \mu_A V - C_0 \bar{R}_0 - a[C_0 \bar{\gamma} + \Delta \bar{R}_0] - a^2 \Delta \bar{\gamma}, \quad C_0 = \mu_A \Lambda, \quad \Delta = \lambda_A - \mu_A \Lambda.$$

Proposition 4 (Stackelberg-optimal pooling rate). *Provided $\Delta \bar{\gamma} > 0$ (second-order condition), the leader's optimal pooling ask rate is*

$$a^{SE} = -\frac{C_0 \bar{\gamma} + \Delta \bar{R}_0}{2\Delta \bar{\gamma}}.$$

Corollary 2 (Bias shifts the pooling rate). *$\partial a^{SE} / \partial \lambda_A$ has the sign of $2\bar{\gamma}a^{SE} - \bar{R}_0$ (by implicit differentiation of the FOC). Under a further regularity condition analogous to Assumption 1 (the residual $\bar{R}(a^{SE})$ stays above half of its $a=0$ level), a^{SE} is strictly decreasing in λ_A : an assistant whose training penalizes clarifying questions more than users actually mind them ($\lambda_A > c_Q$) sets a strictly lower pooling ask rate than an unbiased assistant would.*

Corollary 3 (Non-identification of the two bias channels in the interior regime). *On the interior branch ($\Delta \bar{\gamma} > 0$), substituting $C_0 = \mu_A \Lambda$ and $\Delta = \lambda_A - \mu_A \Lambda$ into Proposition 4 gives*

$$a^{SE} = -\frac{\bar{R}_0}{2\bar{\gamma}} - \frac{1}{2} \left[\frac{1}{\lambda_A / (\mu_A \Lambda) - 1} \right],$$

so a^{SE} depends on the two bias parameters strictly through the scalar ratio λ_A / μ_A (holding Λ , the model primitive, fixed), not through λ_A and μ_A separately. Consequently $\partial a^{SE} / \partial \mu_A = -(\lambda_A / \mu_A) \cdot \partial a^{SE} / \partial \lambda_A$ exactly, and any two parameter pairs on the same ray $\lambda_A / \mu_A = \text{const}$ produce the identical pooling ask rate. A leader that merely discounts downstream welfare ($\mu_A < 1$, $\lambda_A = c_Q$) is therefore observationally indistinguishable, from a^{SE} alone, from a leader with an under-asking friction bias at the corresponding λ_A on the same ray: neither Corollary 2 nor this result identifies which channel is responsible for a low observed ask rate, and a field test would need an instrument independent of a^{SE} itself (e.g. a training-objective measure of λ_A) to separate them.

Remark 4 (Scope of Corollary 3). *The derivation above, and its numerical confirmation, are both restricted to the interior branch $\Delta\bar{\gamma} > 0$. By Corollary 5 below, the unbiased pooling optimum is always a corner, so the ray-invariance result has empirical bite only once bias is already present.*

Corollary 4 (Ray-invariance extends to the corner regime). *On the corner branch ($\Delta\bar{\gamma} \leq 0$), the relevant comparison is $\Pi(1) - \Pi(0)$ rather than the interior stationary point. Substituting $C_0 = \mu_A \Lambda$ and $\Delta = \lambda_A - \mu_A \Lambda$ into that difference gives*

$$\Pi(1) - \Pi(0) = \mu_A \left[\Lambda \bar{R}_0 - \frac{\lambda_A}{\mu_A} (\bar{R}_0 + \bar{\gamma}) \right],$$

so $\mu_A > 0$ again factors out completely and the sign depends only on the scalar ratio λ_A/μ_A , exactly as on the interior branch. The critical ratio is

$$\rho^* = \frac{\Lambda \bar{R}_0}{\bar{R}_0 + \bar{\gamma}},$$

with $a^{SE} = 1$ for $\lambda_A/\mu_A < \rho^$ and $a^{SE} = 0$ for $\lambda_A/\mu_A > \rho^*$. Ray-invariance therefore holds unconditionally across the entire parameter space, interior and corner regimes alike, resolving the open question raised above. A numerical sweep of 11 rays across 10 values of $\mu_A \in [0.1, 1.0]$ (110 evaluations) confirms the analytical threshold exactly and finds zero boundary-crossing violations.*

Corollary 5 (Pooling is a corner solution, not a compromise). *In the unbiased case ($\mu_A = 1, \lambda_A = c_Q$), $\Delta = c_Q - \Lambda$ and $\bar{\gamma} = (\Lambda - c_Q)\mathbb{E}[1/\kappa]$, so*

$$\Delta\bar{\gamma} = -(\Lambda - c_Q)^2 \mathbb{E}[1/\kappa] \leq 0 \quad \text{always.}$$

The second-order condition of Proposition 4 therefore fails in the unbiased case: $\Pi(a)$ is weakly convex on $[0, 1]$, its stationary point is a minimum, and the true optimum is a corner, $a^{SE} \in \{0, 1\}$, not an interior compromise as an earlier draft of this result claimed. Comparing the two corners directly,

$$\Pi(1) - \Pi(0) = (\Lambda - c_Q) \left[\bar{R}_0 - c_Q \mathbb{E}[1/\kappa] \right],$$

so the unbiased pooling optimum is $a^{SE} = 1$ (always ask) when this is positive and $a^{SE} = 0$ (never ask) when negative. Pooling therefore does not average across the population's heterogeneous first-best preferences (Proposition 3); it serves one type's preferred extreme exactly and gives the other type the extreme it least wants—a starker inefficiency than a blended compromise, and one that sharpens rather than weakens the contrast with screening in Section 7: screening's value is not merely “less averaging” but the ability to treat types differently at all. (An interior a^{SE} remains possible away from the unbiased case, specifically when $\lambda_A - \mu_A \Lambda$ and $\Lambda - c_Q$ share a sign, so bias can change not only the level of a^{SE} but whether pooling compromises or picks a winner.)

This last point matters for the comparison in Section 7: Model I has two independent, additive sources of loss relative to first best—the pooling restriction, and bias.

6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). Consider two types $\kappa_L < \kappa_H$ with population shares f_L, f_H , and let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \quad \text{IR}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$.

6.1 Single-crossing

The marginal rate of substitution between m and a along an indifference curve is $\text{MRS}(\kappa) = -\partial U/\partial m/\partial U/\partial a$; using the FOC expressions above, $\partial \text{MRS}/\partial \kappa$ has a consistent sign across the relevant range (higher- κ types have steeper indifference curves in (m, a) -space: they require more compensating ask-service per unit of specification given up). This is the standard Spence–Mirrlees condition and underwrites everything below; we omit the routine but lengthy verification.

6.2 No distortion without bias

Proposition 5 (Efficiency survives private information when the leader is unbiased). *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot; \kappa)$ for every κ), then implementing each type’s own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

Proof. $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot; \kappa)$ by definition, so for any $\kappa' \neq \kappa$, $U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \geq U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$ trivially: no bundle, including another type’s, can give type κ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot; \kappa)$, the leader’s payoff from serving each true type is exactly maximized at that type’s own first best, so this menu is also leader-optimal. $\square$

This is the paper’s sharpest result: unlike classical regulation (Baron–Myerson, Laffont–Tirole), where the principal pays a literal transfer and therefore has an independent reason to dislike conceding rent, here the leader’s payoff can coincide exactly with user welfare, and when it does, incentive compatibility is *free*—private information costs nothing. Distortion in this setting is therefore a pure signature of misalignment, not of asymmetric information per se.

6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot; \kappa)$ (bias present), the leader’s preferred bundle per type, $(m_\kappa^B, a_\kappa^B) = \arg \max_{m,a} \Pi_\kappa(m, a)$ (computable in closed form by the same method as Proposition 3, with Λ, c_Q replaced by their μ_A, λ_A -weighted counterparts), no longer automatically satisfies IC, because Π_κ and the user’s true $U(\cdot; \kappa)$ now rank bundles differently.

Proposition 6 (Downward distortion of the high-cost type’s service). *Under the single-crossing condition and the empirically motivated under-asking bias ($\lambda_A > \mu_A c_Q$, i.e. the leader over-weights the friction of asking relative to true user cost), the leader’s optimal incentive-compatible menu sets $a_L^{SB} = a_L^B$ (no further distortion for the low-cost type, whose biased first-best is already the binding corner $a = 0$) and $a_H^{SB} < a_H^B \leq a_H^{FB} = 1$ (the high-cost type is asked strictly less than even the leader’s own already-biased preference for them), with a_H^{SB} characterized by the standard virtual-surplus first-order condition $\partial \Pi_{\kappa_H}/\partial a_H = \frac{f_L}{f_H} [\kappa_H - \kappa_L] m_H \cdot \partial m_H/\partial a_H$ (rent-adjustment term on the right).*

Proof sketch, corrected. Relax the problem to IC_L alone binding. Unlike classical transfer-based screening (Baron–Myerson, Laffont–Tirole), where the principal’s payoff is directly reduced by the rent conceded to the efficient type—so the standard technique pushes the inefficient type’s IR down to reservation to minimize that rent—here Π_{κ_H} already contains $\mu_A U(\cdot; \kappa_H)$ directly, so the leader’s own payoff from serving the high-cost type *rises* with that type’s true welfare: there is no independent budget motive to depress U_H to $\underline{U}$, and IR_H need not bind. The bundle given to the high-cost type is instead set by its own (biased) unconstrained optimum $(m_H^B, a_H^B) = \arg \max \Pi_{\kappa_H}$, moved only

as far as needed to satisfy IC_L —i.e., only until $U(m_H, a_H; \kappa_L) = U(m_L, a_L; \kappa_L)$ binds—using the envelope representation of the low type’s temptation to mimic, $U(m_H, a_H; \kappa_L) - U(m_H, a_H; \kappa_H) = (\kappa_H - \kappa_L)m_H^2/2$. This still yields $a_H^{SB} < a_H^B$ whenever the unconstrained biased optimum would otherwise violate IC_L , but the resulting first-order condition is not the classical virtual-surplus expression a naive import of Baron–Myerson would suggest; see the remark below. $\square$

Remark 5 (Correction from numerical audit). *An earlier draft of this proof sketch assumed, by direct analogy with Baron–Myerson, that IC_L and IR_H bind while IC_H and IR_L are slack. Solving the full four-variable constrained problem numerically instead finds IC_L binding alone, with IR_H , IC_H , and IR_L slack at representative under-asking-bias parameters, for exactly the reason given in the proof sketch above—and it is the same class of error already corrected once in this model (Proposition 5): a classical mechanism-design result that relies on the principal’s payoff being reduced by conceded rent does not transfer mechanically to a principal whose payoff already contains the agent’s utility. We flag the exact active constraint set as parameter-region-specific rather than proven for all bias magnitudes—for sufficiently extreme bias, Π_{κ_H} and $U(\cdot; \kappa_H)$ could diverge enough that IC_H binds instead, which would change the character of the result. The claim we retain with confidence is qualitative: some constraint binds and produces $a_H^{SB} < a_H^B$ under under-asking bias; the exact active set should be checked numerically for the parameter region of interest rather than assumed from the classical template.*

Remark 6. *The direction in Proposition 6 is specific to the under-asking bias case ($\lambda_A > \mu_{ACQ}$), which we take to be the empirically relevant one (Section 3.3’s motivating story: turn-level satisfaction signals penalize extra clarifying questions more than users actually mind answering them). Under the opposite bias, the direction of distortion can flip; we do not claim a sign that holds for all Π_{κ} .*

7 Comparing the Two Regimes

Corollary 6 (Menus weakly dominate pooling). *The leader’s optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy a is a feasible (degenerate) menu with $a_L = a_H = a$, trivially satisfying IC , so the unconstrained menu optimum is at least as good.*

The more interesting comparison is *where the two regimes’ losses come from*, which Propositions 4–6 let us decompose cleanly:

- **Model I (pooling)** loses welfare from two sources that need not be additive in the same simple sense as in earlier drafts: (i) the pooling restriction itself, present even when $\mu_A=1$, $\lambda_A=c_Q$, where it manifests as a corner solution that fully serves one type and fully fails the other (Corollary 5), and (ii) bias, which shifts a^{SE} (Corollary 2) and can additionally flip pooling from a corner into an interior compromise or vice versa, depending on the sign alignment noted in Corollary 5. The two sources interact rather than simply add.
- **Model II (screening)** loses welfare from bias *only* (Proposition 5): with no bias, screening recovers full efficiency despite private information. With bias, the loss is the rent-extraction-driven downward distortion of Proposition 6, which is a genuinely different mechanism from Model I’s pooling loss (it doesn’t arise from being unable to condition on anything—it arises from not being able to trust what is conditioned on).

Remark 7 (Compounding, testable). *Because Model II’s distortion is triggered by, and scales with, the same bias parameters (μ_A, λ_A vs. c_Q) that drive Model I’s shift in a^{SE} , reducing measured*

assistant bias should improve outcomes for high-cost (terse) users under both regimes, but the welfare gap between menu-conditioned and pooled policies should track population heterogeneity in κ (the spread $\kappa_H - \kappa_L$ and how far F sits on either side of κ^) almost independently of bias. This gives two separable, testable quantities from the same field data: a bias-sensitive term and a heterogeneity-sensitive term.*

8 Testable Predictions

- P1** (Prop. 1, Cor. 1) Holding the assistant’s policy fixed, higher guessing accuracy should be associated with shorter specifications but higher or equal success rates—never both shorter specs and lower success.
- P1’** (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length should flip sign depending on whether $c_Q \geq \Lambda = L(1 - g)$: where clarifying questions are cheap relative to the risk they resolve, users should specify less as ask rate rises (substitution); where questions are relatively costly to answer, users should specify *more* (pre-emption). This is a within-model prediction distinct from P1’s accuracy channel.
- P2** (Prop. 2) The correlation between turn-level satisfaction and eventual task success should weaken, or invert, specifically for under-asking assistants, and can now be benchmarked against the derived a^{SE} of Proposition 4 rather than an assumed ask rate.
- P3** Making ask-rate statistics visible to users should shift specification length toward the sophisticated best response of Section 3.3.
- P4** (Prop. 3) Ask rates should rise with a user’s revealed difficulty specifying detail (terseness), sharply around a threshold κ^* , rather than responding only to model confidence.
- P5** Users with lower self-reported error-catching rates should show declining specification effort over a session (Section 3.2’s dynamic channel).
- P6** (Cor. 3) Under a pooling policy, terse users should be under-served and verbose users over-served relative to *each type’s own* first best, even for a demonstrably unbiased assistant—a gap attributable purely to instrument restriction, distinguishable in data from bias by holding λ_A fixed (e.g., via an A/B test on friction penalties in training) and varying only whether the policy can condition on declared specification.
- P7** (Prop. 6) Even where an assistant *can* condition on declared specification, terse users should still be asked less than the assistant’s own (possibly already-biased) per-type preference would predict, specifically because of the incentive-compatibility constraint—a distortion that should shrink as the population’s cost heterogeneity ($\kappa_H - \kappa_L$) shrinks, holding bias fixed (Remark on compounding).

9 Empirical Strategy (Summary)

Simulation. Instantiate LLM-simulated users with heterogeneous cost types κ interacting with assistants implementing Model I’s derived a^{SE} , Model II’s numerically-solved menu, and naive base-lines (fixed confidence threshold, ignoring both stakes and type). Verify Proposition 3’s threshold numerically, verify Proposition 4’s closed form against a grid search over a , and verify Proposition 5/6 by solving the constrained menu optimization directly and checking that the low type’s

bundle sits at its unconstrained optimum while the high type’s does not, with the gap shrinking as bias is set to zero.

Human study. A between-subjects manipulation of ask policy (pooling vs. menu-conditioned; biased vs. unbiased friction weighting) across repeated coding tasks, measuring specification length and, where feasible, self-reported terseness/verbosity as a proxy for κ , to test P1’, P6, and P7 causally rather than correlationally.

10 Discussion and Limitations

The linear-risk approximation to the conjunctive success function is the main tractability assumption carried through Sections 4–7. A numerical audit re-deriving both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a, g)^{k-m}$ (independently of the linear-risk closed forms, via direct grid search and constrained optimization) finds the corner-solution property of Corollary 5 surviving in every tested configuration across $k \in [3, 20]$, $g \in [0.50, 0.95]$, and the downward-distortion direction of Proposition 6 surviving in the large majority of configurations, with the exceptions concentrated at small k where specification effort saturates at $m = k$ and the asking wedge collapses. This is reassuring for the paper’s two headline qualitative claims. The initial exact-model audit tested only a single bias level for the screening problem, leaving open whether the further finding (Remark following Proposition 6) that IC_H can bind under sufficiently extreme bias was an artifact of the linear-risk approximation. A follow-up sweep of the full four-variable constrained menu problem, solved directly under the exact conjunctive payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ (100 grid points), resolves this: IC_H binds alongside IC_L in 89 of the 100 points, reproducing the same three-regime structure (no binding constraint, IC_L alone, $IC_L + IC_H$ together) found under the linear-risk approximation, with only the boundary between the moderate- and extreme-bias regimes shifting slightly (from $\lambda_A \approx 6.0$ to $\lambda_A \approx 8.0$ at $\mu_A = 1.0$). The extreme-bias reversal is therefore an intrinsic feature of the screening contract space, not an artifact of the linear-risk approximation. Sections 2–3.3 use the exact conjunctive form throughout, so with the above audits the paper’s two halves are now substantially reconciled in exactness level, modulo the small- k saturation caveat noted above.

A related limitation concerns identification of λ_A itself. Every bias-driven prediction (Corollary 2, Proposition 6, and predictions P2, P6, P7) is stated in terms of $\lambda_A > \mu_A c_Q$, but the paper does not currently propose a way to measure λ_A independent of the ask-rate behavior it is meant to explain; backing it out from observed policy alone is circular, and Corollary 3 sharpens this into an exact non-identification result on the interior branch: a^{SE} depends on (μ_A, λ_A) only through their ratio, so a discounting-driven and a friction-driven leader with the same λ_A/μ_A are indistinguishable from pooling behavior alone. An independent handle—for instance, from a training objective or reward specification that explicitly penalizes turn count—would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested non-circularly. The two-type restriction in Model II avoids ironing/continuum-type complications at the cost of generality; a continuum version using the standard Mirrlees technique is the natural next step and would let κ^* from the first-best benchmark interact directly with the screening distortion. We also assume truthful revelation of specified attributes (no cheap-talk vagueness within what is said) and a single leader (no competition among assistants, which would itself be an interesting extension—competition plausibly disciplines bias directly, another testable channel).

Two results in this model (the corner-solution character of unbiased pooling, Corollary 5; and the active constraint set in the biased screening problem, the remark following Proposition 6) were corrected after a companion numerical implementation solved the underlying optimizations

directly rather than trusting a hand-derived proof sketch by analogy with classical mechanism design. Both corrections trace to the same root cause—importing a technique that assumes the principal’s payoff is reduced by conceded surplus, into a setting where the principal’s payoff instead contains the agent’s utility directly—which suggests this is a recurring hazard specific to modeling an assistant whose objective is a (possibly discounted) function of user welfare rather than a literal transfer, and future extensions of this model should treat any imported result from the transfer-based mechanism-design literature as a conjecture to be checked numerically rather than a result to be cited directly.

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